

PLANT INSPECTION REPORT

Inspection & Integrity Department — Field Record

SYNTHETIC DOCUMENT — FICTIONAL DATA FOR THE AEGIS DEMONSTRATION

Report No.:	INS-2026-0588	Equipment Tag:	V-2107
Description:	Amine Flash Drum		
Service:	Rich amine / flash gas		
Service Category:	Corrosive (SOP-INS-014 Cl. 2.2)		
Design Pressure:	7.0 bar(g)	Design Temp.:	90 deg C
Material:	SA-516 Gr.70 Carbon Steel	In Service Since:	18 May 2011
Nominal Thickness:	10.0 mm	t-min:	6.0 mm (INS-014 Cl. 3.2)
Date of Inspection:	18 May 2026	Previous Inspection:	18 May 2024
Inspector:	S. Iyer, Inspection Engineer	Method:	UT survey + internal visual

ULTRASONIC THICKNESS READINGS (mm)

Location	2024	2026	Min. Recorded
Shell course 1 (liquid zone)	7.1	5.8	5.8
Shell course 2 (vapour zone)	9.2	9.0	9.0
Top head	9.5	9.4	9.4
Bottom head	8.6	8.2	8.2
Outlet nozzle N2	8.8	8.3	8.3
Inlet nozzle N1	9.4	9.3	9.3

VISUAL OBSERVATIONS

- Shell course 1: uniform internal wall loss below the normal liquid level. Minimum reading 5.8 mm, below the t-min of 6.0 mm.
- No pitting on shell course 1. No cracking found by magnetic particle examination of the adjacent welds.
- No through-wall defect and no active leak identified.
- Relief valve PSV-2107A bench tested satisfactory on 02 May 2026.
- Vessel is not insulated; external coating in good condition.

INSPECTOR REMARKS

Shell course 1 is below t-min. Vessel isolated at 17:30 on 18/05/2026 and to remain out of service pending engineering assessment. Head of Inspection approved the withdrawal; Plant Manager notified. Recommend replacement of shell course 1 or of the vessel.

Signed: S. Iyer Date: 18/05/2026 Sheet 1 of 1

SYNTHETIC DOCUMENT — FICTIONAL DATA FOR THE AEGIS DEMONSTRATION